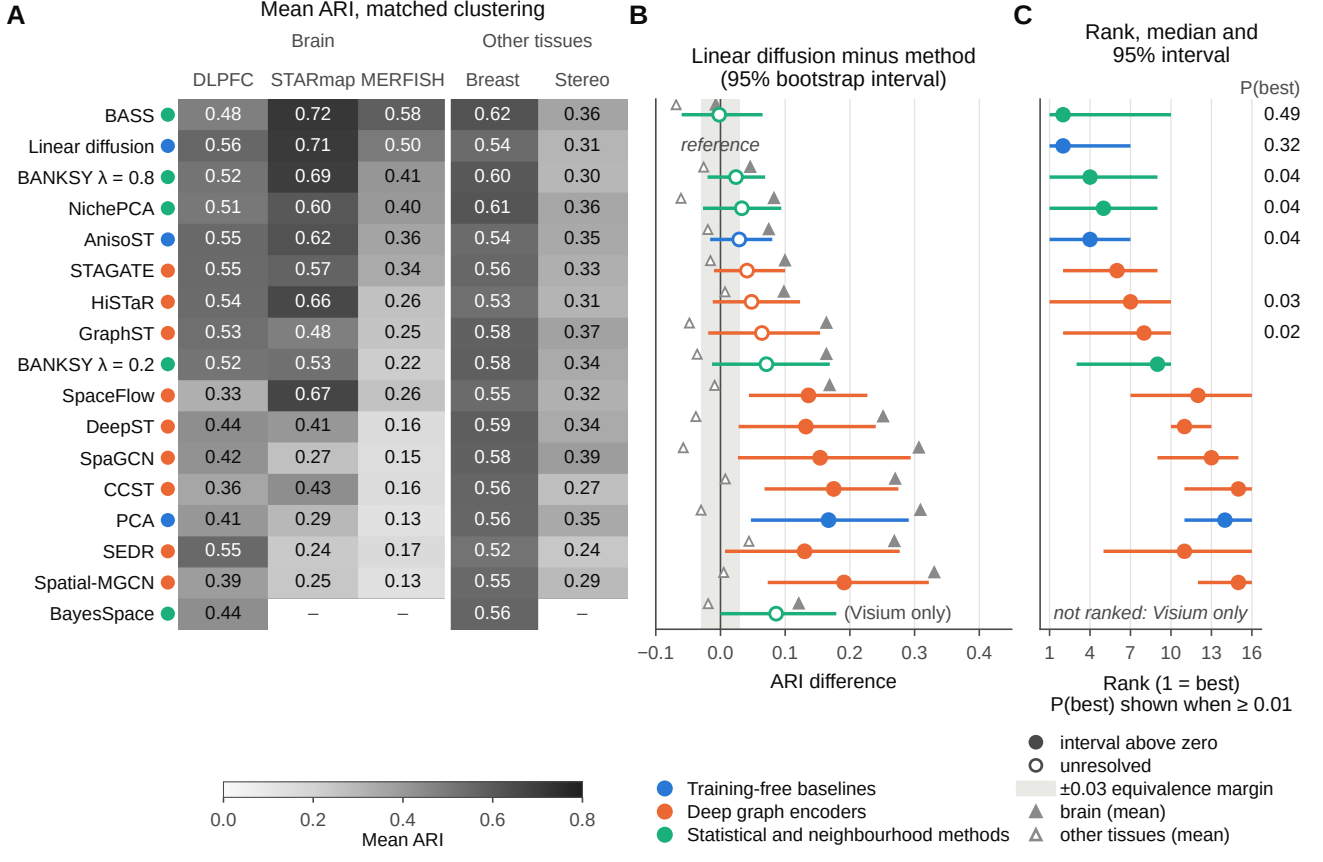

A
Mean ARI, matched clustering
Brain
Other tissues
DLPFC
STARmap
MERFISH
Breast
Stereo
BASS 0.48 0.72 0.58 0.62 0.36
Linear diffusion 0.56 0.71 0.50 0.54 0.31
BANKSY λ = 0.8 0.52 0.69 0.41 0.60 0.30
NichePCA 0.51 0.60 0.40 0.61 0.36
AnisoST 0.55 0.62 0.36 0.54 0.35
STAGATE 0.55 0.57 0.34 0.56 0.33
HiSTaR 0.54 0.66 0.26 0.53 0.31
GraphST 0.53 0.48 0.25 0.58 0.37
BANKSY λ = 0.2 0.52 0.53 0.22 0.58 0.34
SpaceFlow 0.33 0.67 0.26 0.55 0.32
DeepST 0.44 0.41 0.16 0.59 0.34
SpaGCN 0.42 0.27 0.15 0.58 0.39
CCST 0.36 0.43 0.16 0.56 0.27
PCA 0.41 0.29 0.13 0.56 0.35
SEDR 0.55 0.24 0.17 0.52 0.24
Spatial-MGCN 0.39 0.25 0.13 0.55 0.29
BayesSpace 0.44 – – 0.56 –
0.0 0.2 0.4 0.6 0.8
Mean ARI
B
Linear diffusion minus method
(95% bootstrap interval)
reference
(Visium only)
−0.1 0.0 0.1 0.2 0.3 0.4
ARI difference
C
Rank, median and
95% interval
P(best)
0.49
0.32
0.04
0.04
0.04
0.03
0.02
not ranked: Visium only
1 4 7 10 13 16
Rank (1 = best)
P(best) shown when ≥ 0.01
Training-free baselines
Deep graph encoders
Statistical and neighbourhood methods
interval above zero
unresolved
±0.03 equivalence margin
brain (mean)
other tissues (mean)